$P_{\rm orb}\,(M_\star/M_\odot)^{0.5}$ [days]

10^0 10^1 10^2 10^3 10^4 10^5 10^6

fraction of population

0.2

0.0

systems per bin

10^2

80 $M_{\rm J}$

10^1

13 $M_{\rm J}$

10^0

10^{-1}

10^{-2}

10^{-3}

10^{-4}

companion mass m_p [$M_{\rm J}$]

99% contour of each population

J

S

Gaia DR4 baseline

Gaia DR4 `epochalypse` catalog
one-companion high-SNR sub-population

significant peak, period localized

significant peak or acceleration, period not localized

Doppler method (610)

Transit method (623)

Gaia DR4 (Lammers & Winn, 2026):
forecasted orbit characterized (1,938)

Gaia DR4 (Lammers & Winn, 2026):
forecasted orbit detected (27,191, $P_{\rm orb} < 7$ yr)

10^{-2} 10^{-1} 10^0 10^1 10^2

0 10000 20000

semi-major axis a [au]